

Determine whether the upper and lower bound tests indicate k is an upper bound, a lower bound, or neither bound on the real zeros of $f(x)$.

9) $f(x) = x^5 + 5x^4 + 5x^3 + x^2 - 3x - 4$
 $k = 4$

10) $f(x) = 2x^4 - 2x^3 - 6x^2 + 6x$
 $k = -1$

11) $f(x) = 5x^5 - 3x^4 + 5x^3 + 4x^2 + 5x$
 $k = 4$

12) $f(x) = x^5 - 5x^4 + 12x^3 + 5x^2 - 13x$
 $k = -5$

13) $f(x) = x^4 + 4x^3 - 5x^2 - 5x$
 $k = 3$

14) $f(x) = 2x^6 + 5x^5 + 4x^4 + 3x^3 + 4x^2 + 3x - 5$
 $k = -3$

Find an interval in which all real zeros of $f(x)$ lie.

15) $f(x) = 12x^4 - 25x^3 - 8x^2 + 23x - 6$

16) $f(x) = x^5 - 3x^4 - x^3 + 3x^2 + 3x + 1$

17) $f(x) = 3x^4 - 8x^3 - 15x^2 - 4x$

18) $f(x) = 2x^4 - 9x^3 - 7x^2 + 9x + 5$